

Background: Conventional training for chest radiographic diagnosis models primarily relies on structured labels (right bottom table), which are often incomplete, overlooking the rich clinical information embedded in reports created by expert radiologists.

Motivation: Radiology reports provide detailed descriptions of abnormalities, their severity, and contextual findings (highlighted with colors) that are not explicitly captured by categorical labels.

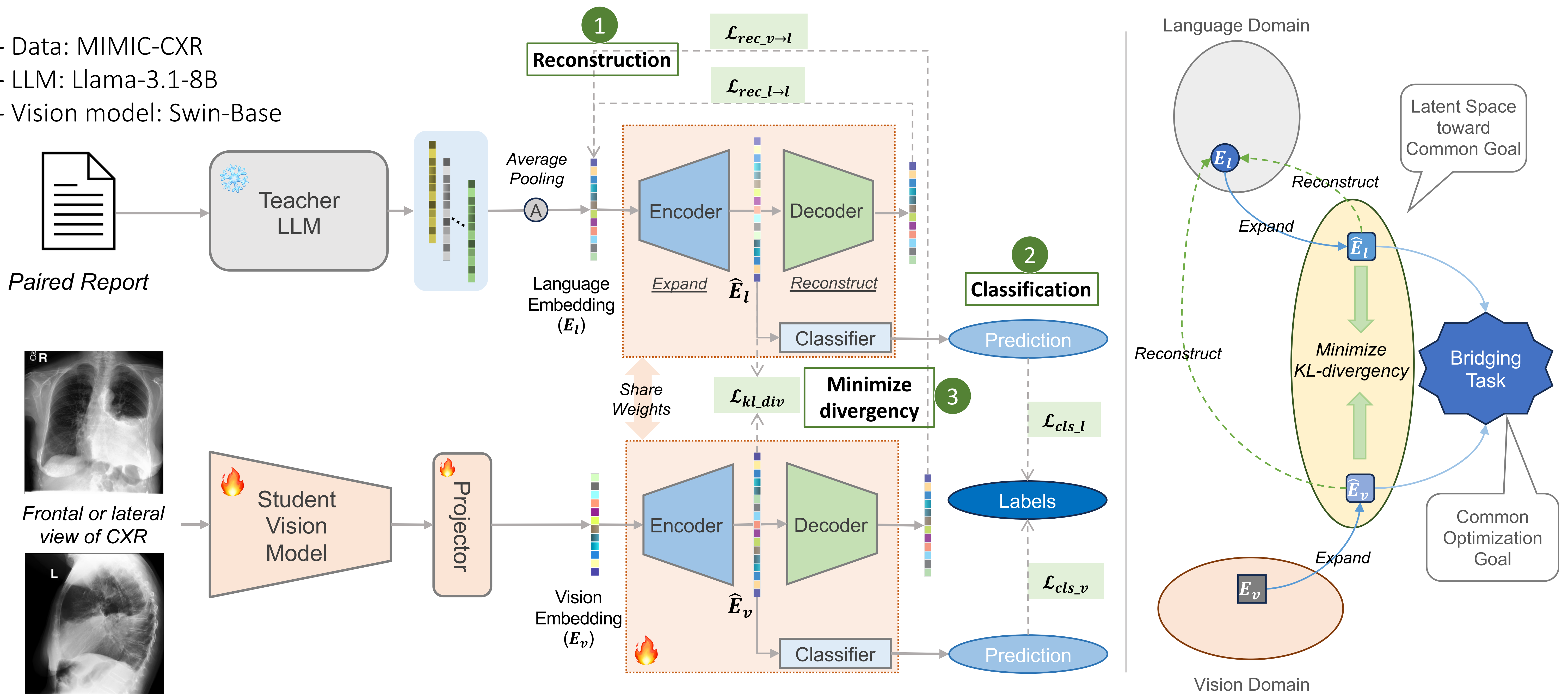
Vision: Distilling essential knowledge contained within radiology reports into the vision model can enhance model's ability to comprehend nuanced diagnostic patterns beyond predefined labels, ultimately improving diagnostic performance.



FINDINGS: There has been mild interval decrease of a still moderate right pleural effusion . There is increased opacification involving the right mid lung zone, likely atelectasis and effusion . There are stable fibrotic changes involving both lungs with left apical scarring compatible with known prior tuberculosis exposure. There are no new focally occurring opacities concerning for pneumonia . There is no evidence of pneumothorax . Cardiomediastinal and hilar contours are stable, with the heart size within the upper limits of normal. Pulmonary vascularity is not increased. There are multiple healed right rib deformities.				
IMPRESSION: Status post right-sided thoracentesis with still a moderate layering right pleural effusion . No evidence of pneumothorax .				
	Atelectasis	Cardiomegaly	Consolidation	Edema
MIMIC labels	Fracture	Lung Lesion	Lung Opacity	No Finding
	Pleural Other	Pneumonia	Pneumothorax	Support Devices
			0.0	Enlarged Cardiomediastinum
				Pleural Effusion 1.0

We aim to develop Medical Vision Foundation Models that distill knowledge from radiology reports via LLMs

- Data: MIMIC-CXR
- LLM: Llama-3.1-8B
- Vision model: Swin-Base



Result 1: SOTA performance cross various thoracic disease diagnostic tasks

Method	Input resolution	Long-tailed diseases				Thoracic diagnoses	Tuberculosis	COVID-19	
		mAUC (%)	mMCC (%)	mAP (%)	mF1 (%)	mAUC (%)	AUC	Accuracy (%)	mMCC
Baseline	224	77.39±0.48	25.19±0.52	28.41±0.62	29.34±0.39	90.35±0.31	93.35±0.77	96.10±0.83	92.28±0.33
MIM-CXR		78.03±0.95	25.25±0.92	28.06±1.01	28.99±0.58	91.71±1.04	95.76±1.79	96.17±0.56	93.88±2.02
Ark-6		82.58±0.12	31.01±0.21	37.03±0.36	33.82±0.26	95.07±0.16	98.98±0.15	97.25±0.40	95.34±0.57
CREED (ours)		82.70±0.10	30.53±0.58	37.05±0.85	33.32±0.70	94.06±0.18	98.28±0.36	97.40±0.34	95.30±0.80
RAD-DINO	518	82.73±0.57	31.10±1.16	37.24±1.90	33.89±1.11	94.74±0.35	98.66±0.08	97.05±0.27	95.58±0.81
CREED (scale-up)	448	83.38±0.15	31.36±0.28	38.57±0.80	34.14±0.38	94.64±0.55	98.92±0.35	97.75±0.64	96.89±0.54

With the best bolded and the second best underlined, a statistical analysis is conducted between the best vs. others, where green-highlighted boxes indicate no statistically significant difference at level $p = 0.05$.

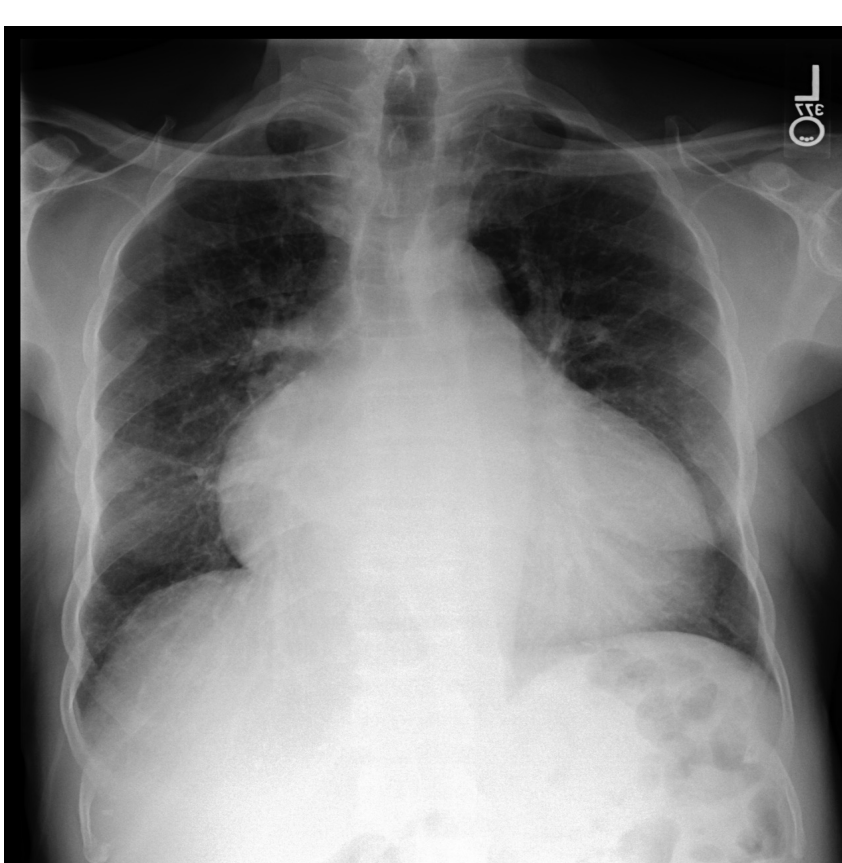
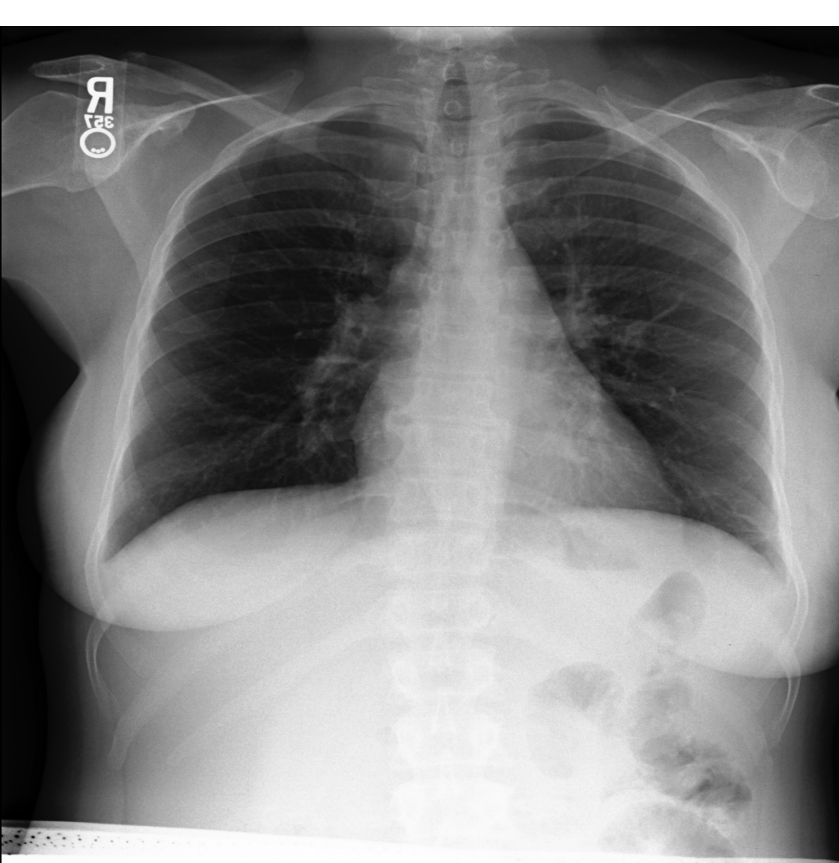
Result 2: Report generation performance using R2GenGPT

Configuration	Parameters	Method	Bleu-1	Bleu-2	Bleu-3	Bleu-4	ROUGE_L	CIDEr
Shallow Alignment	4.2M	R2GenGPT	0.364	0.235	0.161	0.116	0.276	0.135
		CREED (ours)	0.401	0.261	0.182	0.130	0.296	0.195
Deep Alignment	90.9M	R2GenGPT	0.406	0.268	0.186	0.135	0.298	0.268
		CREED (ours)	0.413	0.269	0.189	0.139	0.304	0.262

Result 3: Ablation Study

Configuration	Long-tailed diseases	Tuberculosis
CREED (full model)	82.70±0.10	98.28±0.36
No reconstruction	81.32±0.51(↓1.38)	97.43±0.33(↓0.85)
No classification	80.04±0.75(↓2.66)	96.31±0.45(↓1.97)
No KL-divergence	81.42±0.43(↓1.28)	97.09±0.93(↓1.19)

Result 4: Examples of radiology reports generated by R2GenGPT using CREED

Ground Truth	Generated Report	Ground Truth	Generated Report	
	impression : severe enlargement of the cardiac silhouette unchanged likely cardiomegaly . findings : pa and lateral views of the chest . again seen is severe enlargement of the cardiac silhouette . there is no focal consolidation pleural effusion or pneumothorax . the mediastinal and hilar contours are unchanged . a right central venous catheter has been removed .	impression : cardiomegaly with mild pulmonary vascular congestion . findings : pa and lateral views of the chest provided . there is no focal consolidation pleural effusion or pneumothorax . cardiac silhouette is mildly enlarged . mediastinal contours are stable . no acute osseous abnormalities .		impression : no acute cardiopulmonary process . findings : cardiac mediastinal and hilar contours are normal . pulmonary vasculature is normal . no focal consolidation pleural effusion or pneumothorax is present . there are no acute osseous abnormalities .
			impression : no acute cardiopulmonary process . findings : pa and lateral views of the chest were obtained . there is no focal consolidation pleural effusion or pneumothorax . the cardiomeastinal silhouette is normal . imaged osseous structures are intact . no free air below the right hemidiaphragm is seen .	